TechNova Enterprise Network

Design, Implementation, Security and Verification
PROJECT DOCUMENTATION

1. Project Overview

This project documents a simulated enterprise network for TechNova, consisting of a Head Office (HQ), a Branch Office and an ISP-facing router. It demonstrates departmental VLANs, centralized network services, inter-VLAN routing, static routing, secure remote management, access control, switch port security and NAT/PAT.

Platform: Cisco Packet Tracer. The external destination is simulated using an ISP router loopback interface; this is not a connection to the public Internet.

2. Network Components

Routers: HQ-R1, ISP-R2 and BRANCH-R3.

Switches: CORE-SW1, HR-IT-SW2, FIN-SALES-SW3, SERVER-SW4 and BRANCH-SW5.

Servers: DHCP-DNS-SERVER (192.168.50.10) and APPLICATION-SERVER (192.168.50.11).

End devices: HR, IT, Finance, Sales and Administration PCs, plus BRANCH-PC1.

3. VLAN and IP Addressing Plan

VLAN 10 — HR: 192.168.10.0/24; gateway 192.168.10.1.

VLAN 20 — IT: 192.168.20.0/24; gateway 192.168.20.1.

VLAN 30 — Finance: 192.168.30.0/24; gateway 192.168.30.1.

VLAN 40 — Sales: 192.168.40.0/24; gateway 192.168.40.1.

VLAN 50 — Servers: 192.168.50.0/24; gateway 192.168.50.1.

VLAN 99 — Management: 192.168.99.0/24; gateway 192.168.99.1.

VLAN 60 — Branch: 192.168.60.0/24; gateway 192.168.60.1.

4. Router Links and Static Routing

HQ-R1 G0/0: 10.0.0.5/30; ISP-R2 G0/0: 10.0.0.6/30.

HQ-R1 G0/2: 10.0.0.1/30; BRANCH-R3 G0/0: 10.0.0.2/30.

HQ-R1 has a static route to 192.168.60.0/24 via 10.0.0.2 and a default route via 10.0.0.6.

BRANCH-R3 has a default route via 10.0.0.1. ISP-R2 has a return route for 192.168.0.0/16 via 10.0.0.5.

5. Switching and Inter-VLAN Routing

Departmental VLANs were created on the HQ switches. Access ports were assigned to the relevant VLANs and uplinks were configured as trunks.

CORE-SW1 was configured as the STP root bridge for VLANs 10, 20, 30, 40, 50 and 99.

HQ-R1 uses router-on-a-stick on G0/1 with 802.1Q subinterfaces for VLANs 10, 20, 30, 40, 50 and 99. BRANCH-R3 connects to Branch VLAN 60 through its routed interface.

6. DHCP, DNS and Application Services

DHCP-DNS-SERVER uses 192.168.50.10/24, gateway 192.168.50.1 and DNS address 192.168.50.10. DHCP pools were configured for HQ VLANs, with DHCP relay on HQ-R1 subinterfaces using ip helper-address 192.168.50.10.

BRANCH-R3 provides DHCP for 192.168.60.0/24, with 192.168.60.1 as the default gateway and 192.168.50.10 as DNS.

APPLICATION-SERVER uses 192.168.50.11/24 with gateway 192.168.50.1. HTTP and HTTPS were enabled. The DNS A record www.xyz.local points to 192.168.50.11.

7. Secure Remote Management — SSH

SSH version 2 was configured on the three routers and four HQ switches. Local user authentication and RSA keys were configured, and VTY lines were set to accept SSH. Remote SSH sessions were tested from end devices.

Lab credentials are not included in this document. Use unique credentials in any real environment.

8. Access Control List — Department Segmentation

A named extended ACL, BLOCK_HR_TO_FINANCE, was configured on HQ-R1. It denies IP traffic sourced from 192.168.10.0/24 to 192.168.30.0/24 and permits other IP traffic.

The ACL is applied outbound on HQ-R1 G0/1.30. Testing showed HR-to-Finance traffic blocked while HR access to the application server remained successful. ACL match counters were observed.

9. Switch Port Security

Port security was configured on access ports using a maximum of one MAC address, sticky MAC learning and violation mode restrict.

HR-IT-SW2: Fa0/1–Fa0/4. FIN-SALES-SW3: Fa0/1–Fa0/4. SERVER-SW4: Fa0/1–Fa0/2 and Fa0/4–Fa0/5. BRANCH-SW5: Fa0/2.

Trunk uplinks and router-facing ports were excluded. Verification used show port-security and show port-security address. Learned MAC entries depend on traffic being generated by connected endpoints.

10. NAT/PAT

HQ-R1 was configured with G0/0 as the NAT outside interface. The HQ VLAN subinterfaces and G0/2 (Branch-facing link) were configured as NAT inside interfaces.

Standard ACL 1 permits 192.168.0.0/16. PAT overload uses the G0/0 interface address (10.0.0.5) as the shared outside address.

ISP-R2 Loopback0 was configured as 203.0.113.1/32 as a simulated external test destination. NAT translations showed Branch-PC1 (192.168.60.21) and HR-PC1 (192.168.10.100) translated to 10.0.0.5.

11. Verification and Test Evidence

Reported successful checks include inter-VLAN gateway connectivity, Branch-to-HQ reachability, DNS resolution for www.xyz.local, HTTP access to the application server, ACL enforcement, SSH access and NAT/PAT translations.

Useful commands: show vlan brief; show interfaces trunk; show spanning-tree root; show ip interface brief; show ip route; show ip ssh; show access-lists; show port-security; show port-security address; show ip nat statistics; show ip nat translations.

Add the actual Packet Tracer screenshots to the portfolio repository. Screenshots should show the complete topology and relevant command output or client tests.

12. Conclusion and Learning Outcomes

The project provided practical experience with enterprise network segmentation, switching, router-on-a-stick, DHCP relay, DNS and HTTP services, static routing, SSH administration, ACL-based traffic control, switch port security and

PAT.

This is a Packet Tracer simulation. It demonstrates configuration and verification in a lab environment; it does not establish production deployment experience or real public-Internet connectivity.

Appendix: Suggested Evidence Files

01_Network_Topology.png; 02_VLAN_Configuration.png; 03_Trunk_Configuration.png; 04_Routing_Table.png; 05_DHCP_Verification.png; 06_DNS_Verification.png; 07_HTTP_Server.png; 08_SSH_Verification.png; 09_ACL_Security.png; 10_Port_Security.png; 11_NAT_PAT.png; 12_End_to_End_Testing.png.

Packet Tracer file: TechNova_Enterprise_Network.pkt. Keep the .pkt file and evidence screenshots alongside this documentation.